

Table Splitting in Blocks

Whole Table

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

...

Split into Blocks

1	2	3	4
5	6	7	8

B1

9	10	11	12
13	14	15	16

B2

17	18	19	20
21	22	23	24

B3

25	26	27	28
29	30	31	32

B4

Sparse Order Routing Attention

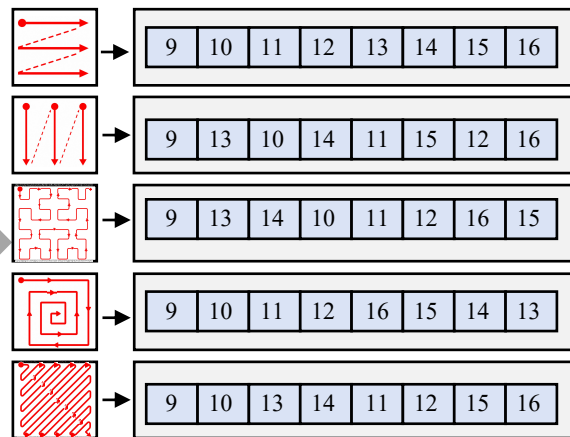
Stage 1: Warm-up

row-major only

Stage 2: MoE

routing enabled

A Certain Layer



Self-Attention

row output
(base)

column output
(routed)

hilbert output
(routed)

spiral output
(routed)

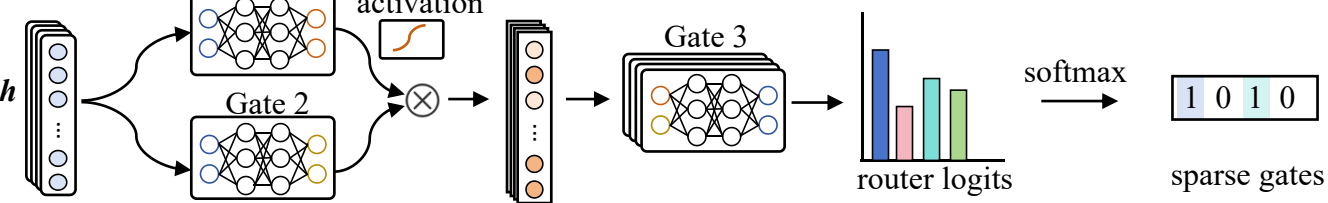
snake output
(routed)

Router

A Certain
Head

Attention
output

Router



Based shared order



Routed orders



Data flow



Residual sum